Greenhouse Management
Validation + Verification

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1 Validation

Validation is the assurance that a product, service, or system meets the needs of the customer and other identified stakeholders.

In this document, I intend to work on revising the web-based greenhouse dashboard to reflect the feedback received - applying updates and improvements towards a better result or IT product.

This project aims to build a web-based platform to help farmers monitor their crops. By centralizing critical environmental data, i.e., temperature, humidity, soil moisture, and light, into an accessible dashboard, the system enables users to make data-driven decisions that result in harvest optimization while reducing production costs.

Figure 1 shows the final product as presented in my realization document. I went on asking feedback about this from the teacher Raja and she looked at it. She said it was good. However, if I could make the graph bigger to cover

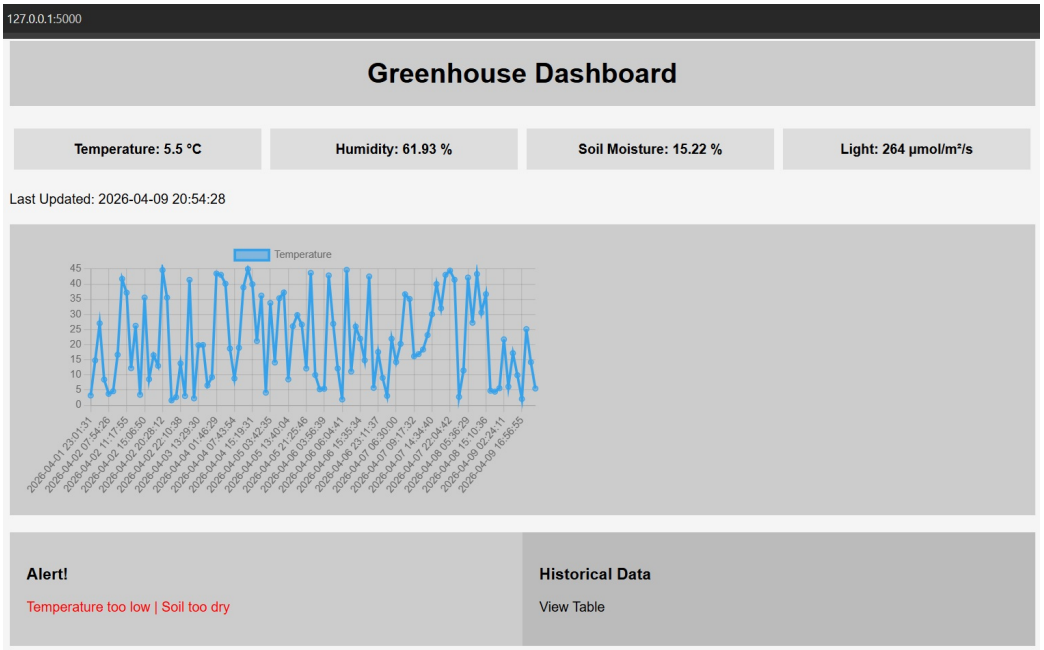


Figure 1: Greenhouse Dashboard Main Page

the entire container reserved for it, then it would be even better.

That was a good suggestion. I then worked on it, changed the dimensions of the graph in my template and it looked better as seen in Figure 2.

She also asked for the X and Y labels on the chart to be made more visible and readable, and I thought that was actually a great idea. I have been meaning to put more colors on my dashboard to make it look more lively. I told myself that I will add more colors to it after making sure that the requirements are working. Which I did, now it would be wonderful if I could make it more attractive than how it looked in Figure 2.

I did not just work on making the font more visible, I also worked on making the chart a bit more colorful - different line colors per chart as seen in Figure 3. This looks great, however, this is not how I wanted it to be. Because here, the cards stopped working. When you clicked on them, they are not responding. If you would like to see the charts, then you will have to click on the legends - those small colorful rectangles.

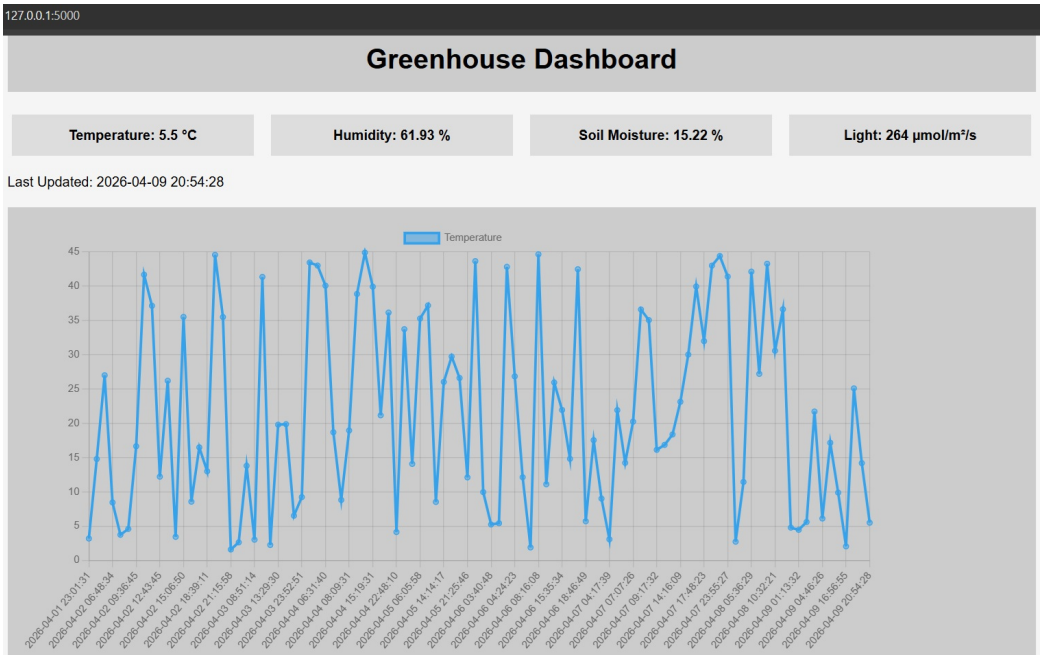


Figure 2: The Chart made Wider

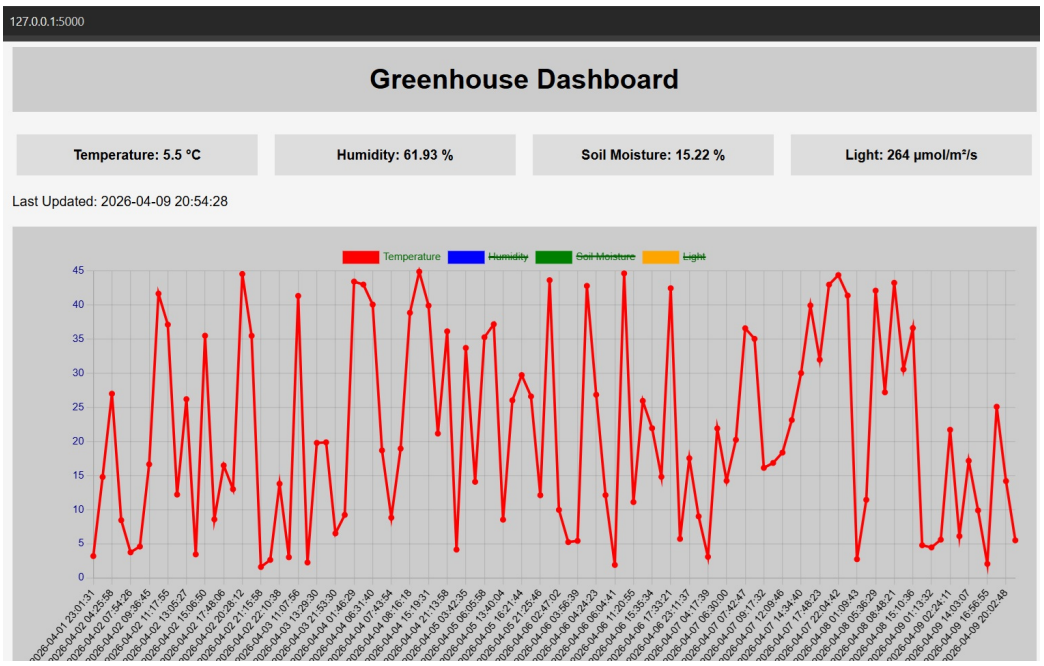
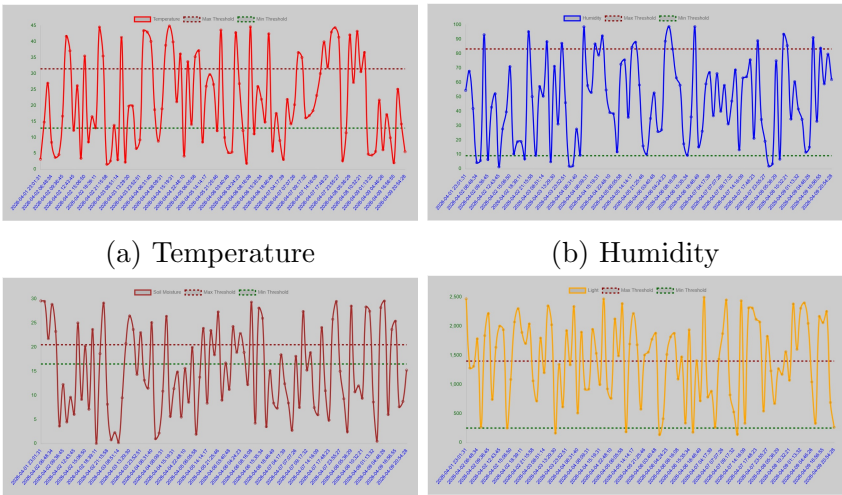


Figure 3: One Chart - Multiple Legends

What I did, I worked on making sure that the cards works again and also added the threshold indicator lines to show which numbers fall within the normal threshold as shown in Figure 4.



(a) Temperature (b) Humidity

(c) Soil Moisture (d) Light (PAR)

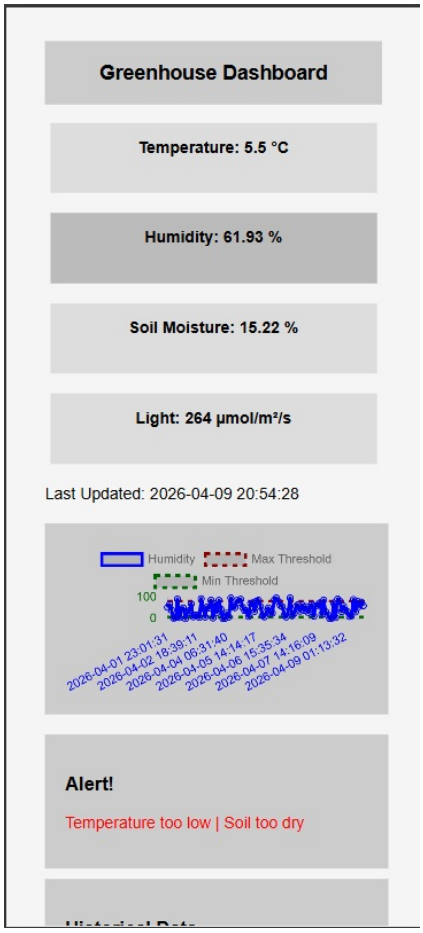
Figure 4: Different Charts with Threshold Indicators

After working on adding colors to the chart, and upon reviewing the requirements I had enumerated in the design of this project, I realized that I was missing the mobile view feature of this dashboard where users can also view the dashboard on their phones. With the help of AI [1], I was able to learn how it can be done and applied it to my project as shown in Figure 5.

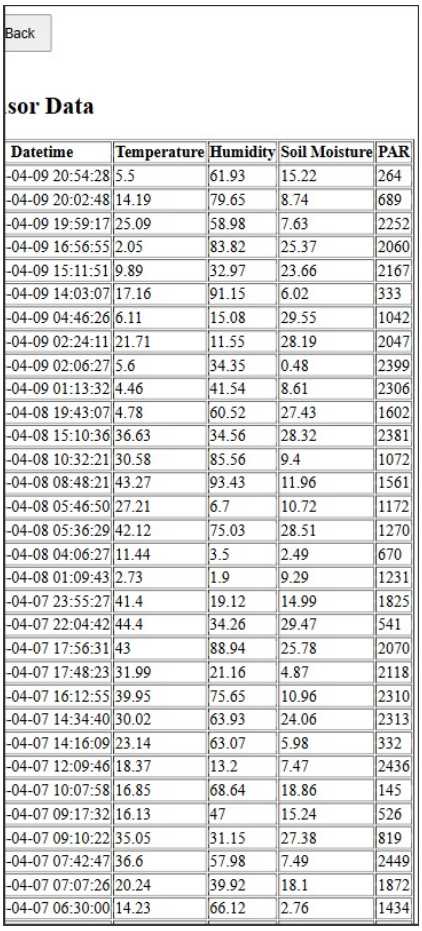
This mobile view works exactly like the web view where users can click on the different cards like Temperature, Humidity, Soil Moisture, and Light. However, the graph looks so small and if given more time, I might be able to make it look nicer.

But for now, this is how the dashboard looks like on the desktop and mobile view.

[1] Greenhouse IT Integration



(a) Main Page



Back

sor Data

Datetime	Temperature	Humidity	Soil Moisture	PAR
-04-09 20:54:28	5.5	61.93	15.22	264
-04-09 20:02:48	14.19	79.65	8.74	689
-04-09 19:59:17	25.09	58.98	7.63	2252
-04-09 16:56:55	2.05	83.82	25.37	2060
-04-09 15:11:51	9.89	32.97	23.66	2167
-04-09 14:03:07	17.16	91.15	6.02	333
-04-09 04:46:26	6.11	15.08	29.55	1042
-04-09 02:24:11	21.71	11.55	28.19	2047
-04-09 02:06:27	5.6	34.35	0.48	2399
-04-09 01:13:32	4.46	41.54	8.61	2306
-04-08 19:43:07	4.78	60.52	27.43	1602
-04-08 15:10:36	36.63	34.56	28.32	2381
-04-08 10:32:21	30.58	85.56	9.4	1072
-04-08 08:48:21	43.27	93.43	11.96	1561
-04-08 05:46:50	27.21	6.7	10.72	1172
-04-08 05:36:29	42.12	75.03	28.51	1270
-04-08 04:06:27	11.44	3.5	2.49	670
-04-08 01:09:43	2.73	1.9	9.29	1231
-04-07 23:55:27	41.4	19.12	14.99	1825
-04-07 22:04:42	44.4	34.26	29.47	541
-04-07 17:56:31	43	88.94	25.78	2070
-04-07 17:48:23	31.99	21.16	4.87	2118
-04-07 16:12:55	39.95	75.65	10.96	2310
-04-07 14:34:40	30.02	63.93	24.06	2313
-04-07 14:16:09	23.14	63.07	5.98	332
-04-07 12:09:46	18.37	13.2	7.47	2436
-04-07 10:07:58	16.85	68.64	18.86	145
-04-07 09:17:32	16.13	47	15.24	526
-04-07 09:10:22	35.05	31.15	27.38	819
-04-07 07:42:47	36.6	57.98	7.49	2449
-04-07 07:07:26	20.24	39.92	18.1	1872
-04-07 06:30:00	14.23	66.12	2.76	1434

(b) Table Page

Figure 5: Different Charts with Threshold Indicators

2 Verification

Verification is the evaluation of whether or not a product, service, or system complies with a regulation, requirement, specification, or imposed condition.

In the analysis and advice document of the project, I listed certain functional and non-functional requirements that the product of this project should satisfy upon its completion. I enumerated those requirements below as a checklist to show whether the final product met those requirements or not.

- ☑ Data Collection - the system shall collect data such as humidity, light, temperature, and soil moisture, and shall support real-time or simulated sensor data input.

  I used a mock data on this project to simulate sensor data for temperature, humidity, soil moisture, and light.

- ☑ Data Storage - the system shall store collected data and save the data with timestamps for tracking overtime.

  The mock data also include data and time for the timestamps and it was stored in a database via SQL.

- ☑ Data Processing - the system shall clean and preprocess data and shall organize data for analysis.

  This has been done via Python using sqlite3.

- ☑ Data Analysis - the system shall perform time-series analysis to show trends, shall detect anomalies, and shall generate alerts when thresholds are exceeded.

  This has also been done together with processing the data via ssqlite3 in Python.

- ☑ User Interface (Dashboard) - the system shall display real-time data on a dashboard. shall show graphs, and shall allow users to view historical data.

  Python has used with flask, html, css and javascript to build the dashboard. They have the cards for the sensor data, has the chart, and users can also view the historical data in a table format.

- ☑ Alerts and Notification - the system shall notify users when conditions exceeds safe limits and alerts shall be visible on the dashboard.

The dashboard has an alert box that warns the user whenever the data reached outside the threshold.

- ☐ Performance - the system should update data in real-time and should respond quickly to user actions.

  This was not met because the I used a mock data to simulate the sensor data for temperature, humidity, soil moisture, light. Thus, they were not real-time but they do respond quickly to user actions.

- ☑ Usability - the system should have a simple and user-friendly interface, and the dashboard should be easy to understand for non-technical users.

- ☑ Reliability - the system should operate consistently without frequent failures, and should handle errors without crashing.

- ☑ Maintainability - the system should be easy to maintain and modify. Code should be easy to organized and documented.

  I have added markdown notes in the template to label what a certain group of codes does hopefully to help the user understand how they work.

- ☑ Compatibility - the system should work on common web browsers, and the dashboard should be accessible on both desktop and mobile devices.

3 Future Scope

Looking ahead, the Greenhouse Management project will grow from a simple dashboard into a fully smart and independent system. The next step involves using machine learning models that can look at data to predict exactly what a plant needs before a problem even starts. Instead of just showing information, the greenhouse will use advanced automation systems to take action on its own, like automatically turning on water or adjusting the lights.

To help even larger farms, the project will expand into large-scale sensor network installations, allowing farmers to monitor thousands of plants at once with high precision. The ultimate vision is a full-scale commercial deployment, bringing these powerful tools to the global market to make farming easier, cheaper, and more productive for everyone.

4 Conclusion

This greenhouse management project shows how technology can make farming easier and more efficient. By collecting data such as temperature, humidity, soil moisture, and light, the system helps farmers understand what is happening inside the greenhouse at any time.

The use of a database, data analysis, and a web dashboard allows data monitoring and quick decision-making. Features like charts and alerts helps detect problems early, which can protect crops and improve productivity.

Overall, this project demonstrates that using simple IT solutions can improve sustainability, reduce manual work, and increase the success of greenhouse farming.